

— GEN-GEN SPEAKER CONSISTENCY

Voice Design TTS

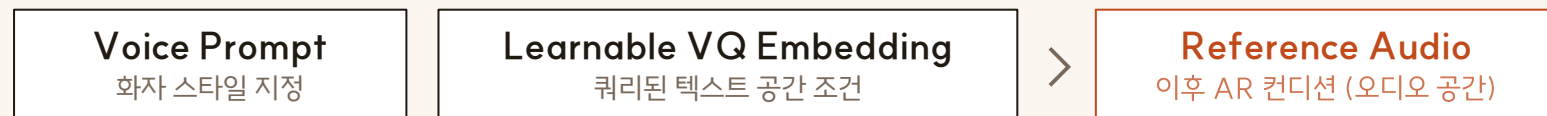
화자 일관성 개선 논문

추가 진행사항

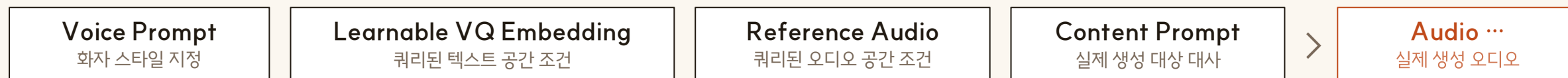
2026.07.20. 미팅

VQ임베딩 토큰 사용 (COS 0.423/WER 0.135)

[1] TEACHER-FORCED RECONSTRUCTION으로 QUERY만 학습



[2] AR 연속 생성



앵커 축 (학습 없음)						
A1	앵커 길이 tiny(2tok)	0.336	0.124	3.32	✓	너무 짧으면 하락
A2	길이 short(11)	0.353	0.122	3.43	✓	
A3	길이 long(25)	0.368	0.119	3.52	✓	
A4	길이 xlong(47)	0.352	0.117	3.47	✓	포화-미세 하락
A5	greedy anchor (seed+greedy)	0.408	0.131	3.29	✓	앵커 결정화 소폭 이득
A6	multi-ref M=1	0.394	0.124	3.34	✓	=베이스
A7	multi-ref M=4	0.345	0.121	3.44	✓	다중 앵커 오히려 하락
A8	multi-ref M=8 (short)	0.348	0.156	3.39	▲	
A9	temperature 0.9	0.408	0.131	3.29	✓	
A10	temperature 0.6	0.396	0.123	3.35	✓	
A11	temperature 0.3	0.696	0.194	3.13	✗ 무효	샘플링 제한 → 제약 위반
Learnable Global Query (파라미터화)						
G1	raw (k=4, codec-CE만)	0.392	0.604	2.72	✗	off-manifold, WER 붕괴
G2	raw k=4 + sub-talker loss	0.404	0.422	3.39	✗	WER 여전히 붕괴
G3	global residual 벡터	0.350	0.117	3.43	✓	WER-safe but 이득 없음
G4	softmax(vocab 블록 결합)	0.479	0.314	3.27	✗	높은 COS, WER 붕괴
G5	VQ (k=8, ref_text 불일치)	0.512	0.408	3.02	✗	최고 COS지만 WER 붕괴
G6	linear	0.299	0.394	2.53	✗	전부 나쁨
VQ Learnable Global Query in Reftext (k-ablation)						

But, WER 하락은 리뷰 단계에서 지적 가능성이 큰 상황
=> WER 유지할 수 있는 디테일한 조합 필요

#	실험	COS(sim)	WER	UTMOS	WER-safe	비고
베이스라인 / 메커니즘						
B1	two-pass sequential (cached, voice prompt 없음)	0.402	~0.12	—	✓	원본 순차
B2	two-pass batched (cached)	0.346	0.122	3.47	✓	배치화로 하락
B3	two-pass batched (no-cache)	0.269	0.122	3.46	✓	fresh ref
B4	single-pass (cached, voice prompt 보임)	0.348	0.127	3.39	✓	
B5	single-pass (no-cache)	0.266	0.127	3.40	✓	
B6	single-pass no-cache fixedanchor	0.394	0.124	3.34	✓	WER-safe 기준 베이스
Fixed Reference Text Prompt 템플릿						
F1	desc_raw	0.391	0.128	3.34	✓	
F2	natural	0.405	0.123	3.42	✓	템플릿 중 최고
F3	read_aloud	0.385	0.125	3.45	✓	
F4	recording	0.398	0.129	3.48	✓	
앵커 축 (학습 없음)						
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A2	길이 short(11)	0.353	0.122	3.43	✓	
A3	길이 long(25)	0.368	0.119	3.52	✓	
A4	길이 xlong(47)	0.352	0.117	3.47	✓	포화·미세 하락
A5	greedy anchor (seed+greedy)	0.408	0.131	3.29	✓	앵커 결정화 소폭 이득

앵커 측 (학습 없음)						
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A2	길이 short(11)	0.353	0.122	3.43	✓	
A3	길이 long(25)	0.368	0.119	3.52	✓	
A4	길이 xlong(47)	0.352	0.117	3.47	✓	포화·미세 하락
A5	greedy anchor (seed+greedy)	0.408	0.131	3.29	✓	앵커 결정화 소폭 이득
A6	multi-ref M=1	0.394	0.124	3.34	✓	=베이스
A7	multi-ref M=4	0.345	0.121	3.44	✓	다중 앵커 오히려 하락
A8	multi-ref M=8 (short)	0.348	0.156	3.39	⚠	
A9	temperature 0.9	0.408	0.131	3.29	✓	
A10	temperature 0.6	0.396	0.123	3.35	✓	
A11	temperature 0.3	0.696	0.194	3.13	✗ 무효	샘플링 제한 → 제약 위반
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G6	linear	0.299	0.394	2.53	✗	전부 나뻐

VQ Learnable Global Query in Reftext (k-ablation)

ref_text 일치

VQ Learnable Global Query in Reftext (k-ablation)

— ref_text 일치

V1	vqr k=1	0.318	0.157	3.37	⚠	
V2	vqr k=4	0.347	0.131	3.25	✓	
V3	vqr k=8	0.382	0.137	3.04	✓	
V4	vqr k=16	0.347	0.132	3.26	✓	
V5 ★	vqr k=32	0.423	0.135	3.20	✓	WER-safe SOTA

Learnable Per-sample Query / Speaker-Embed
Generator

P1	persona→query generator (gen_k16)	—	—	—	✗	collapse
P2	speaker-embed gen (MLP, CE)	collapse	—	—	✗	per-persona cos 0.96–1.0
P3	+ ECAPA anti-collapse head	미미	—	—	✗	$\lambda=10$ 에도 약함
P4	MiniLM persona encoder 주입	0.217	—	—	✗	collapse는 풀렸으나 주입 실패
P5	단순 Linear (frozen backbone, CE)	collapse	—	—	✗	div $\approx$ 1.0 (아래 §7)

모델 Fine-tuning / SFT

S1	LoRA reconstruction (초기, 레이아웃 오류)	0.377	0.967	0.67	✗	garbage
S2	reconstruction SFT (streaming, light 150step)	0.364	0.122	3.33	✓	WER 유지, COS 이득 없음(I)
S3	ICL 연속화 SFT (v1 aggressive / v2 gentle)	—	broke	—	✗	텍스트 무시

V2	vqr k=4	0.347	0.131	3.25	✓	
V3	vqr k=8	0.382	0.137	3.04	✓	
V4	vqr k=16	0.347	0.132	3.26	✓	
V5 ★	vqr k=32	0.423	0.135	3.20	✓	WER-safe SOTA
Learnable Per-sample Query / Speaker-Embed Generator						
P1	persona→query generator (gen_k16)	—	—	—	✗	collapse
P2	speaker-embed gen (MLP, CE)	collapse	—	—	✗	per-persona cos 0.96–1.0
P3	+ ECAPA anti-collapse head	미미	—	—	✗	$\lambda=10$ 에도 약함
P4	MiniLM persona encoder 주입	0.217	—	—	✗	collapse는 풀렸으나 주입 실패
P5	단순 Linear (frozen backbone, CE)	collapse	—	—	✗	div $\approx$ 1.0 (아래 §7)
모델 Fine-tuning / SFT						
S1	LoRA reconstruction (초기, 레이아웃 오류)	0.377	0.967	0.67	✗	garbage
S2	reconstruction SFT (streaming, light 150step)	0.364	0.122	3.33	✓	WER 유지, COS 이득 없음(!)
S3	ICL 연속화 SFT (v1 aggressive / v2 gentle)	—	broke	—	✗	텍스트 무시
S4	joint (backbone LoRA + Linear, CE)	collapse	broke	—	✗	div 0.9987
S5	joint + contrastive spread (LAM=5)	0.242	0.266	2.00	✗	distinct code지만 off-manifold

VQ임베딩 토큰 사용 + 임베딩 공간 변형

QwenTTS는 Token 임베딩 공간 차원(D_text)과 Talker의 시퀀스 모델링 공간 차원(2048)이 다름

새로운 시도: 미리 토큰 공간에서 시퀀스 모델링 공간에 투영



기존 방식



```
projected
param # 학습 벡터, (K, 2048) ← projected 공간
PROJ = text_projection(E) # 어휘 전체를 미리 투영 (V, 2048)
ids = nearest(param, PROJ) # 2048 공간에서 최근접
q_used = PROJ[ids] # 그대로 사용 (이미 투영됨)

raw
raw_query # 학습 벡터, (K, D_text) ← raw 공간에
ids = nearest(raw_query, E) # D_text 공간에서 최근접
q_used = text_projection(E[ids]) # 쓸 때 투영

핵심 차이: 비선형 warp 때문에 최근접 토큰이 달라짐
```

공간	수치
Baseline	COS 0.2139/WER 0.1186
기존 SOTA	COS 0.423/WER 0.135
기존 방식 다른 시드	0.415/0.180 · 0.433 /0.149 · 0.403/0.150
미리 토큰 공간 투영	0.398/ 0.130 · 0.392/0.143 · 0.414/0.155
미리 토큰 공간 투영 + Cosine LR	0.414/0.162 · 0.424/0.235

Learning 단계에서 SEED 민감성 발견됨
=> 리포트 방안 결정 필요 (멀티 시드?)

InstructTTEval with Qwen3Omni

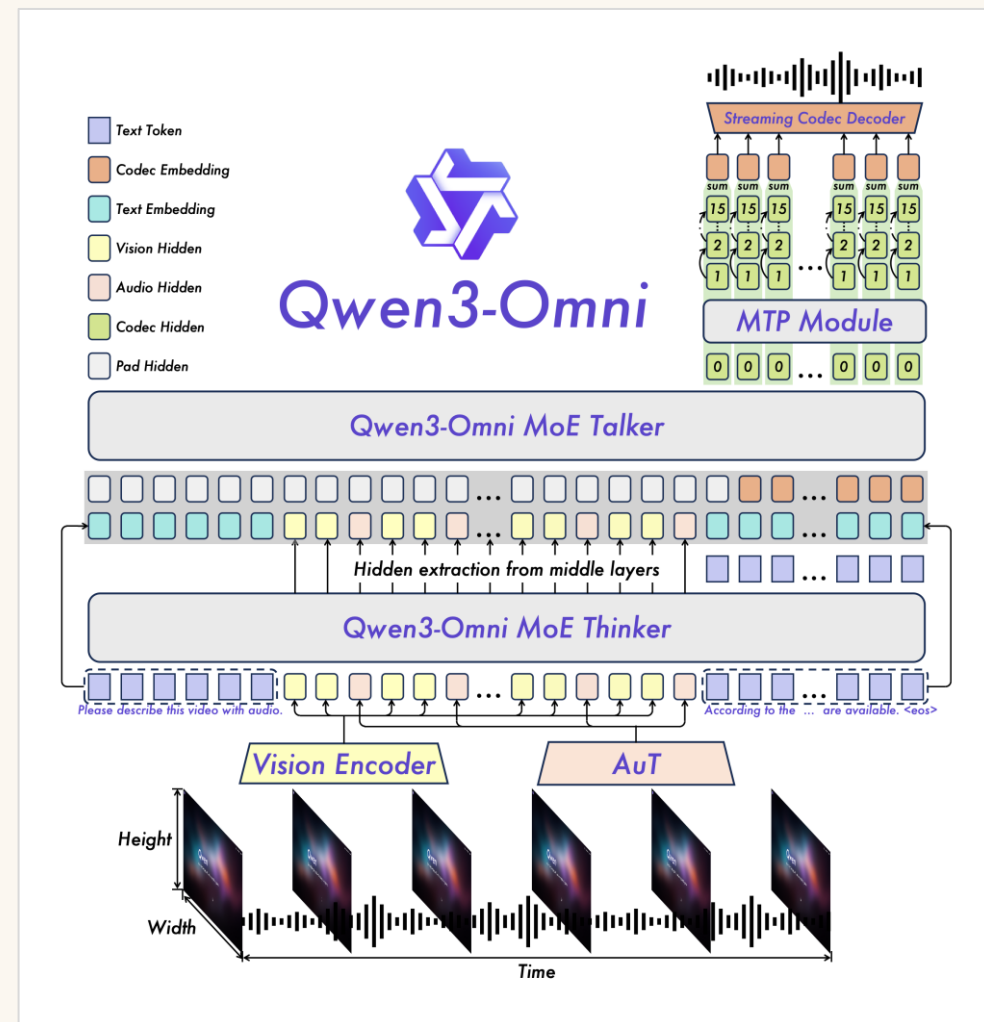
벤치마크 정보

모델: Qwen3-Omni-30B-A3B-Thinking-INT4

- BF16 체크포인트는 VRAM 초과로 로드 불가
- FP8 체크포인트는 사용 가능해야 하나, 미리 양자화된 가중치들이 문제가 있어 추론 실패
- BNB를 이용한 INT4로 시도하여 진행된 작업 (성능 신뢰도가 낮아 좀 더 확인 필요)

추론 시간

- INT4 기준 6시간~7시간



InstructTTEval with Qwen3Ommi (INT4, 3000개)

메서드	APS (1000)	DSD (1000)	RP (1000)
Baseline (VD)	96.2%	92.3%	65.4%
Two-stage (VD 후 VC)	94.2%	89.1%	69.0%
Fixed Anchor	97.1%	91.3%	65.2%
Vector	running...	running...	running...

Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

1 st Given Name Surname <i>dept. name of organization (of Aff.)</i> <i>name of organization (of Aff.)</i> City, Country email address or ORCID	2 nd Given Name Surname <i>dept. name of organization (of Aff.)</i> <i>name of organization (of Aff.)</i> City, Country email address or ORCID	3 rd Given Name Surname <i>dept. name of organization (of Aff.)</i> <i>name of organization (of Aff.)</i> City, Country email address or ORCID
4 th Given Name Surname <i>dept. name of organization (of Aff.)</i> <i>name of organization (of Aff.)</i> City, Country email address or ORCID	5 th Given Name Surname <i>dept. name of organization (of Aff.)</i> <i>name of organization (of Aff.)</i> City, Country email address or ORCID	6 th Given Name Surname <i>dept. name of organization (of Aff.)</i> <i>name of organization (of Aff.)</i> City, Country email address or ORCID

Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.], *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—Component formatting, style, styling insert

I. INTRODUCTION

This document is a model and instructions for L^AT_EX. Please observe the conference page limits.

II. EASE OF USE

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

III. PREPARE YOUR PAPER BEFORE STYLING

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—L^AT_EX will do that for you.

Identify applicable funding agency here. If none, delete this.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m²” or “webers per square meter”, not “webers/m²”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm³”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not

“Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

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Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in L^AT_EX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

BIB_TE_X does not work by magic. It doesn’t get the bibliographic data from thin air but from .bib files. If you use BIB_TE_X to produce a bibliography you must send the .bib files.

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Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.

- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
- The prefix “non-” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

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“Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

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- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings of heads, and organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.

Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

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REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first . . .”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

REFERENCES

[1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, “Title of paper if known,” unpublished.

[5] R. Nicole, “Title of paper with only first word capitalized,” J. Name Stand. Abbrev., in press.

[6] Y. Yonezawa, M. Hirano, K. Ohno, and Y. Taniyama, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” IEEE Trans. Magn., Japan vol. 2, pt. 2, pp. 740–741, August 1982, Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[7] M. Young, The Technical Writer’s Handbook. Mill Valley, CA: University Science, 1989.

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CONCLUSION

RESULTS

REFERENCES

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